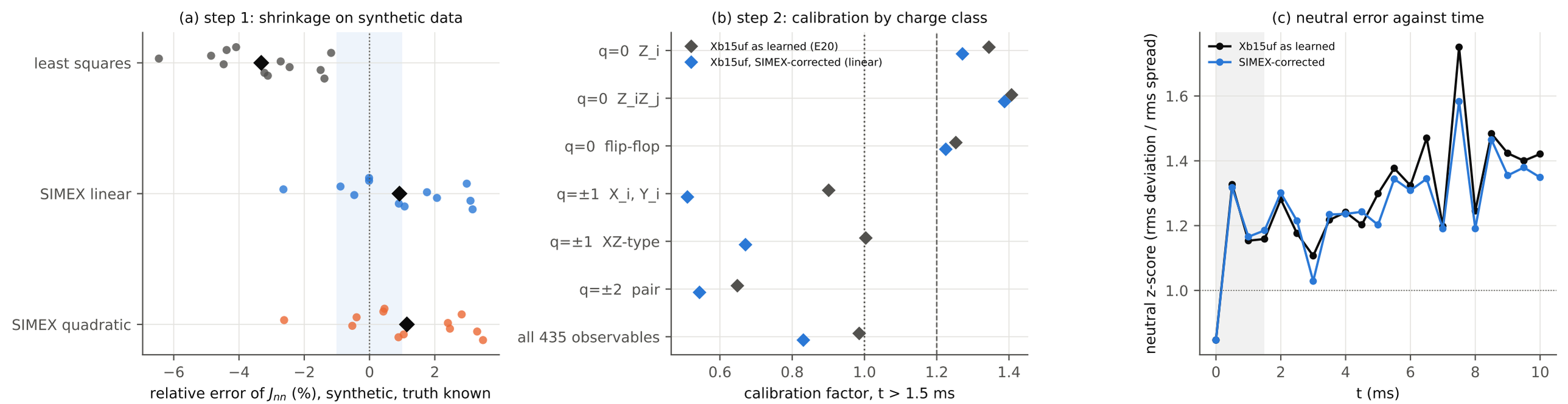
(a) step 1: shrinkage on synthetic data
least squares
SIMEX linear
SIMEX quadratic
−6
−4
−2
0
2
relative error of J_{nn} (%), synthetic, truth known
(b) step 2: calibration by charge class
Xb15uf as learned (E20)
Xb15uf, SIMEX-corrected (linear)
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
q=±1 X_i, Y_i
q=±1 XZ-type
q=±2 pair
all 435 observables
0.6
0.8
1.0
1.2
1.4
calibration factor, t > 1.5 ms
(c) neutral error against time
Xb15uf as learned
SIMEX-corrected
neutral z-score (rms deviation / rms spread)
1.0
1.2
1.4
1.6
0
2
4
6
8
10
t (ms)